TYPE 5: Sum and Difference of 2 Cubes

Example: Multiply the following out and simplify:

a. $(x+y)(x^2-xy+y^2)$

x^3+y^3

b. $(2a+3b)(4a^2-6ab+9b^2)$

$8a^3+27b^3$

c. $(a-b)(a^2+ab+b^2)$

a^3-b^3

d. $(x-5)(x^2+5x+25)$

x^3-125

What do you notice about the final results? Refer to MAS page 26 to make notes on factorising cubes.

Exercise 10

1. x^3+y^3 $(x+y)(x^2-xy+y^2)$
2. $27x^3+64y^3$ $(3x+4y)(9x^2-12xy+16y^2)$
3. x^3-y^3 $(x-y)(x^2+xy+y^2)$
4. $3P^3-3q^3$ $3(p-q)(p^2+pq+q^2)$
5. a^3-27b^3 $(a-3b)(a^2+3ab+9b^2)$
6. x^9-125y^3 $(x^3-5y)(x^6+5x^3y+25y^2)$
7. $8a^3+b^3$ $(2a+b)(4a^2-2ab+b^2)$
8. $(a+2b)^3-(a-2b)^3$
9. $125a^3-64b^3$ $(5a-4b)(25a^2+20ab+16b^2)$
10. $16y^3-432$ $16(y^3-27)$ $16(y-3)(y^2+3y+9)$

Further practice: MAS Page 27 Exercise 8

8) $((a+2b)-(a-2b))((a+2b)^2+(a+2b)(a-2b)+(a-2b)^2)$
$= 4b(a^2+4ab+4b^2+a^2-4b^2+a^2-4ab+4b^2)$
$= 4b(3a^2+4b^2)$

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